

CHAPTER – XII

DATABASE

DATABASE:

- A database is an organized collection of data stored electronically so it can be easily accessed, managed, and updated.
- In Java, a database is accessed using JDBC (Java Database Connectivity), which allows Java programs to:
 - Store data
 - Retrieve data
 - Update data
 - Delete data
- Java itself does not store data permanently; databases like MySQL handle permanent storage.

MySQL:

- An open-source Relational Database Management System (RDBMS)
- Uses SQL (Structured Query Language)
- Stores data in the form of tables (rows and columns)

PARTS OF MySQL:

- **Database** - A container that holds tables
- **Table** - Stores data in rows and columns
- **Rows (Records)** - Each row represents one entity
- **Columns (Fields)** - Each column represents an attribute
- **Primary key** - Uniquely identifies each record and Cannot be NULL
- **Foreign key** - Links two tables and Maintains relationships

MySQL OPERATORS:

1. ARITHMETIC OPERATOR

Operator	Meaning
+	Addition
-	Subtraction
*	Multiplication
/	Division
%	Modulus

2. COMPARISON OPERATOR

Operator	Meaning
=	Equal
!= or <>	Not equal
>	Greater than
<	Less than
>=	Greater than or equal
<=	Less than or equal

3. LOGICAL OPERATOR

Operator	Meaning
AND	Both conditions true
OR	Any condition true
NOT	Reverses condition

MySQL CLAUSES:

- Clauses help filter, sort, and group data.
 - **WHERE** : Filters records - **SELECT * FROM students WHERE marks>50;**
 - **SELECT** : Retrieves data - **SELECT name, marks FROM students;**

SQL COMMAND CATEGORIES:

- **DDL** - Data Definition Language
- **DML** - Data Manipulation Language
- **DQL** - Data Query Language
- **TCL** - Transaction Control Language
- **DCL** - Data Control Language

DDL – DATA DEFINITION LANGUAGE:

- DDL commands are used to define, create, modify, and delete the structure of database objects such as tables, databases, and indexes.
- **Commands:**
 - **CREATE** - Creates a new database or table
 - **ALTER** - Modifies an existing table structure
 - **DROP** - Deletes a table or database
 - **TRUNCATE** - Removes all records from a table (structure remains)
 - **RENAME** - Renames a database object

DML – DATA MANIPULATION LANGUAGE:

- DML commands are used to insert, update, delete, and modify data stored in tables.
- **Commands :**
 - **INSERT** - Adds new records to a table
 - **UPDATE** - Modifies existing records
 - **DELETE** - Removes records from a table

DQL – DATA QUERY LANGUAGE:

- DQL commands are used to retrieve data from the database.
- **Commands :**
 - **SELECT** - Fetches data from one or more tables

TCL – TRANSACTION CONTROL LANGUAGE:

- TCL commands are used to manage transactions and control changes made by DML commands.
- **Commands :**
 - **COMMIT** - Saves changes permanently
 - **ROLLBACK** - Undoes changes
 - **SAVEPOINT** - Sets a point to roll back to later

DCL – DATA CONTROL LANGUAGE:

- DCL commands are used to control access and permissions in the database.
- **Commands :**
 - **GRANT** - Gives permissions to users
 - **REVOKE** - Removes permissions from users